Volume of Cones, Pyramids and Spheres

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

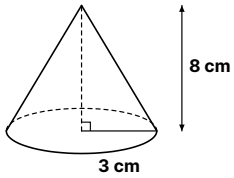
KEY WORDS cone *(koeko)* · pyramid *(koeko tapawhā)* · sphere *(pōro)* · hemisphere *(haurua pōro)*

A Cones ●○○

Remember: A cone is one third of the cylinder it fits inside: Volume $= \frac{1}{3}\pi r^2 h$. Use the perpendicular height, not the slant. Give answers to 1 decimal place.

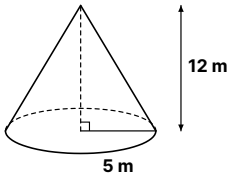
1. Find the volume of each cone.

a)



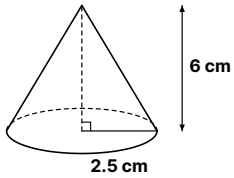
_____

b)



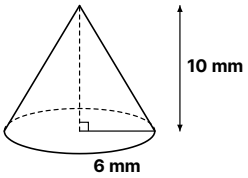
_____

c)



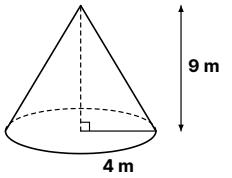
_____

d)



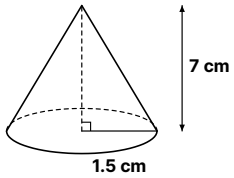
_____

e)



_____

f)



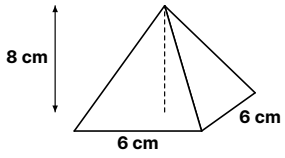
_____

B Pyramids ●●○

Remember: A pyramid is one third of the prism with the same base and height: Volume $= \frac{1}{3}$ × base area × height.

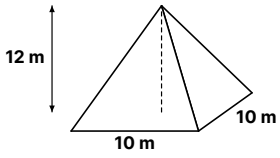
2. Find the volume of each pyramid.

a)



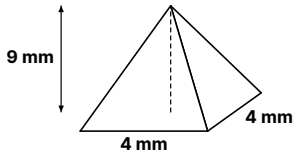
_____

b)



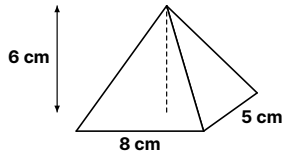
_____

c)



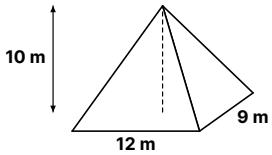
_____

d)



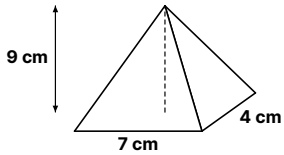
_____

e)



_____

f)



_____

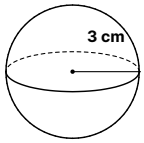
© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C Spheres and hemispheres ●●○

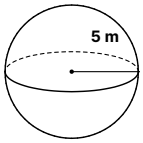
Remember: Volume of a sphere $= \frac{4}{3}\pi r^3$. Cube the radius first. A hemisphere is half of that.

3. Find the volume of each solid, to 1 decimal place.

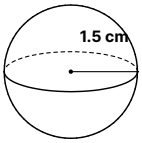
a)



b)



c)



d) A hemisphere (half a sphere) with radius 10 mm

e) A hemisphere (half a sphere) with radius 4 cm

f) A hemisphere (half a sphere) with radius 2.5 m

D Working backwards ●●●

Remember: Put the volume into the formula and solve for the missing length. For a sphere you end with r^3: take the cube root.

4. Find the missing length.

a) A cone has a volume of 150 cm³ and a radius of 4 cm. Find its height, to 1 decimal place.

b) A square-based pyramid has a volume of 400 m³ and base edges of 10 m. Find its height.

c) A sphere has a volume of 500 cm³. Find its radius, to 1 decimal place.

d) A cone has a volume of 1,200 mm³ and a radius of 9 mm. Find its height, to 1 decimal place.

e) A square-based pyramid has a volume of 96 cm³ and base edges of 6 cm. Find its height.

f) A sphere has a volume of 3,000 mm³. Find its radius, to 1 decimal place.

E Problems ●●●

5. An ice cream is a cone of radius 3 cm and height 10 cm, with a hemisphere of ice cream of the same radius on top. Find the total volume, to 1 decimal place.

6. Sand is tipped into a conical pile with a radius of 2.5 m and a height of 1.8 m. Find the volume of sand, to 2 decimal places.

7. A ball of diameter 20 cm is packed in a cube-shaped box with edges of 20 cm. Find the volume of empty space in the box, to the nearest cm³.

8. A roof is a square-based pyramid with base edges of 8 m and a height of 3 m. Find the volume of the roof space.

★ Extension

9. A cone, a hemisphere and a cylinder all have the same radius r, and the cone and cylinder have height r. Show that their volumes are in the ratio 1 : 2 : 3.

10. A sphere of radius 6 cm is melted and recast as a cone with the same radius. Find the height of the cone.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Volume of Cones, Pyramids and Spheres

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

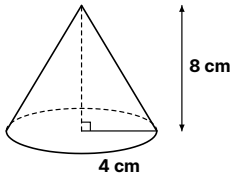
KEY WORDS cone *(koeko)* · pyramid *(koeko tapawhā)* · sphere *(pōro)* · hemisphere *(haurua pōro)*

A Cones ●○○

Remember: A cone is one third of the cylinder it fits inside: Volume $= \frac{1}{3}\pi r^2 h$. Use the perpendicular height, not the slant. Give answers to 1 decimal place.

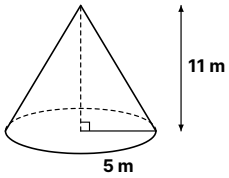
1. Find the volume of each cone.

a)



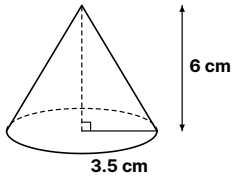
_____

b)



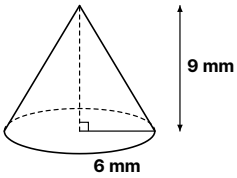
_____

c)



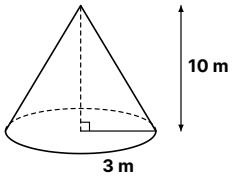
_____

d)



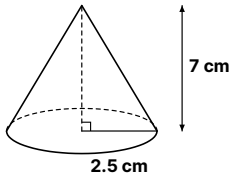
_____

e)



_____

f)



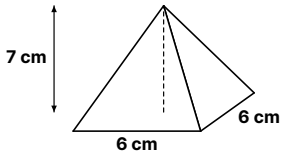
_____

B Pyramids ●●○

Remember: A pyramid is one third of the prism with the same base and height: Volume $= \frac{1}{3}$ × base area × height.

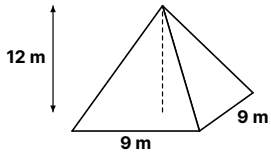
2. Find the volume of each pyramid.

a)



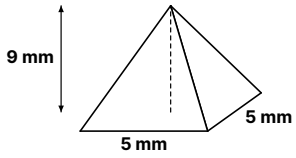
_____

b)



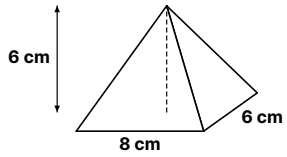
_____

c)



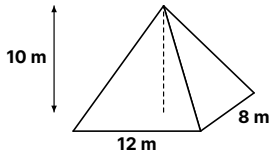
_____

d)



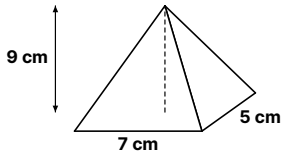
_____

e)



_____

f)



_____

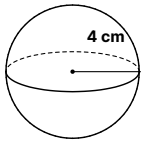
© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

C Spheres and hemispheres ●●○

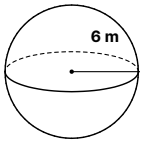
> **Remember:** Volume of a sphere $= \frac{4}{3}\pi r^3$. Cube the radius first. A hemisphere is half of that.

3. Find the volume of each solid, to 1 decimal place.

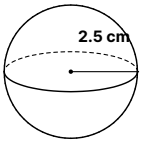
a)



b)



c)



d) A hemisphere (half a sphere) with radius 9 mm

e) A hemisphere (half a sphere) with radius 5 cm

f) A hemisphere (half a sphere) with radius 3.5 m

D Working backwards ●●●

> **Remember:** Put the volume into the formula and solve for the missing length. For a sphere you end with r^3: take the cube root.

4. Find the missing length.

a) A cone has a volume of 200 cm³ and a radius of 5 cm. Find its height, to 1 decimal place.

b) A square-based pyramid has a volume of 324 m³ and base edges of 9 m. Find its height.

c) A sphere has a volume of 800 cm³. Find its radius, to 1 decimal place.

d) A cone has a volume of 1,500 mm³ and a radius of 10 mm. Find its height, to 1 decimal place.

e) A square-based pyramid has a volume of 128 cm³ and base edges of 8 cm. Find its height.

f) A sphere has a volume of 4,000 mm³. Find its radius, to 1 decimal place.

E Problems ●●●

5. An ice cream is a cone of radius 3.5 cm and height 11 cm, with a hemisphere of ice cream of the same radius on top. Find the total volume, to 1 decimal place.

6. Sand is tipped into a conical pile with a radius of 3 m and a height of 2.1 m. Find the volume of sand, to 2 decimal places.

7. A ball of diameter 24 cm is packed in a cube-shaped box with edges of 24 cm. Find the volume of empty space in the box, to the nearest cm³.

8. A roof is a square-based pyramid with base edges of 9 m and a height of 4 m. Find the volume of the roof space.

★ Extension

9. A cone, a hemisphere and a cylinder all have the same radius r, and the cone and cylinder have height r. Show that their volumes are in the ratio 1 : 2 : 3.

10. A sphere of radius 9 cm is melted and recast as a cone with the same radius. Find the height of the cone.

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.